The Thermodynamic Arrow of Time and Quantum Relational Clocks in Chronodynamic Relativity: Unifying Cosmic Expansion, Total Field Energy Dissipation, and the Dynamic Acceleration Horizon

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The nature of time presents two fundamental unresolved problems in modern theoretical physics: the thermodynamic arrow of time (the microscopic time-reversibility of dynamical laws versus macroscopic entropy growth) and the canonical quantum "problem of time" ($\hat{H}\Psi = 0$ in Wheeler-DeWitt quantum geometrodynamics). In this paper, we demonstrate that within Chronodynamic Relativity, both crises resolve naturally when physical time is identified as the manifestation of the non-equilibrium energy dissipation rate of an all-permeating space-density scalar field $\rho_s(\vec{x}, t)$ governed by a conserved total field capacity E_{total}. We prove that the cosmic expansion rate $H(t) \equiv \dot{R}/R = 1/t$ is identically the fractional energy dissipation rate of the background space field: $H(t) = \frac{1}{E_{\text{field}}}|\frac{\mathrm{d}E_{\text{field}}}{\mathrm{d}t}| = \frac{1}{\rho_s}|\frac{\mathrm{d}\rho_s}{\mathrm{d}t}|$. Under this epoch-independent formulation: (1) macroscopic entropy production ($\mathrm{d}S/\mathrm{d}t \geq 0$) and the cosmological arrow of time are microscopic consequences of the monotonic relaxation of the primordial space-density gradient; (2) generalized Page-Wootters relational quantum clocks coupled to the space-fluid vacuum state yield atomic transition frequencies that scale with local field density $\omega_0(\rho_s) \propto \sqrt{\rho_s}$, deriving the cosmological metric clock-rate scaling $\eta(z) = (1+z)^{-n/2}$ ($n = 3/4\pi \approx 0.2387$); and (3) the galactic acceleration scale $a_c(t) = \frac{c}{4\pi}H(t) = \frac{c}{4\pi t}$ represents the exact dynamic boundary where local gravitational field acceleration balances the cosmic energy dissipation rate of the background medium ($g_N \sim c \cdot |\dot{\rho}_s|/\rho_s$), producing a smooth asymptotic crossover into the algebraic root gravity regime $g_{\text{eff}}(r,t) = \frac{1}{2}[g_N + \sqrt{g_N^2 + 4g_N a_c(t)}]$ without dark matter particles or piecewise threshold triggers. We connect these results to non-equilibrium fluctuation theorems (Jarzynski, Crooks), environmental decoherence rates, and quasilocal black hole apparent horizon relaxation timescales ($\tau_{\text{relax}} \sim 1/H(t)$), proposing laboratory tests using sub-millimeter optical lattice clock interferometry.

I. INTRODUCTION

Time occupies a deeply paradoxical status across the foundations of physics:

1. **The Thermodynamic Arrow of Time:** The fundamental microscopic equations of classical mechanics, electromagnetism, and quantum theory are strictly time-reversal symmetric ($\mathcal{T}$-symmetric). Yet, macroscopic phenomena exhibit a strict, non-reversible directionality dictated by the Second Law of Thermodynamics:

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{T}\frac{\mathrm{d}E_{\text{diss}}}{\mathrm{d}t} \geq 0. \tag{1}$$

   Standard cosmology attempts to resolve this via the "Past Hypothesis" (the assumption of an ultra-low entropy initial boundary state at the Big Bang [1, 2]) without providing a dynamic field-theoretic mechanism for why such a boundary state existed or how it connects to the metric expansion of spacetime.

2. **The Quantum Problem of Time:** In canonical quantum gravity, applying Dirac quantization to generally covariant General Relativity yields the Wheeler-DeWitt equation [3]:

$$\hat{H}\Psi = 0. \tag{2}$$

   Because the total Hamiltonian is a pure constraint, the quantum state of the universe Ψ is static; time disappears from the fundamental ontology. To recover dynamical evolution, relational quantum clock frameworks (such as the Page-Wootters mechanism [4] and thermal time hypotheses [5, 6]) introduce internal subsystem clocks, but leave the physical coupling between atomic clock frequencies and cosmic expansion undefined.

3. **The Cosmic-Galactic Acceleration Coincidence:** In galactic astrophysics, the transition from Newtonian dynamics to flat rotation curves occurs at a universal acceleration scale [7]:

$$a_0 \approx 1.635 \times 10^{-10}\ \mathrm{m/s^2} \sim cH_0 \cdot \frac{3}{4\pi}. \tag{3}$$

   In standard ΛCDM, this relation is regarded as an unexplained cosmic coincidence between localized baryonic dynamics and global dark energy expansion.

In this paper, we demonstrate that within **Chronodynamic Relativity (CR)** [8], these three foundational

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problems are not distinct anomalies, but three manifestations of a single physical phenomenon: **time is the physical rate of energy dissipation of the cosmic space-density field**.

Unlike phenomenological Modified Newtonian Dynamics (MOND) [9] (which posits an ad hoc static acceleration scale), kinematic Milne cosmology [10] (which assumes an empty universe without gravitational dynamics), or varying speed of light models [11] (which alter coordinate c and break Lorentz invariance), Chronodynamic Relativity derives cosmic expansion, metric clock lapse $\eta(z)$, and the dynamic acceleration horizon $a_c(t) = \frac{c}{4\pi}H(t)$ directly from first principles: the continuous non-equilibrium dilution of a conserved cosmic field capacity E_{total}.

II. TOTAL FIELD ENERGY AND COSMIC EXPANSION AS FIELD DISSIPATION

In Chronodynamic Relativity, physical space is a flat Euclidean 3-manifold ($\mathbb{R}^3$, $\gamma_{ij} = \delta_{ij}$) filled with a dynamical scalar space-density field $\rho_s(\vec{x}, t)$. The metric line element is governed by local field density:

$$\mathrm{d}s^2 = -c^2\eta^2(\vec{x}, t)\mathrm{d}t^2 + \delta_{ij}\mathrm{d}x^i\mathrm{d}x^j, \quad \eta(\vec{x}, t) \equiv \sqrt{\frac{\rho_s(\vec{x}, t)}{\rho_{\text{ref}}}}, \tag{4}$$

where ρ_{ref} is a baseline reference density.

Let the total primordial space-field energy contained within the causal cosmic horizon be a conserved capacity E_{total}:

$$E_{\text{total}} = \int_{V(t)} \rho_s(\vec{x}, t)\, c^2\, \mathrm{d}^3x = \text{constant}. \tag{5}$$

Because the causal boundary propagates linearly at the speed of light:

$$R(t) = c \cdot t \implies \dot{R}(t) = c, \quad \ddot{R}(t) = 0, \tag{6}$$

the background space density $\rho_s(t)$ represents the geometric dilution of this conserved energy capacity across the expanding causal horizon $R(t)$:

$$\rho_s(t) = \frac{E_{\text{total}}}{4\pi c^3 t} \propto \frac{1}{t}. \tag{7}$$

The fractional rate of energy dissipation of the cosmic space field is:

$$\Gamma_{\text{field}}(t) \equiv \frac{1}{E_{\text{field}}(t)}\left|\frac{\mathrm{d}E_{\text{field}}}{\mathrm{d}t}\right| = \frac{1}{\rho_s(t)}\left|\frac{\mathrm{d}\rho_s}{\mathrm{d}t}\right| = \frac{1}{t}. \tag{8}$$

Comparing this directly with the kinematic Hubble expansion rate $H(t) \equiv \dot{R}/R$, we establish the foundational identity:

$$H(t) \equiv \frac{\dot{R}(t)}{R(t)} = \frac{1}{E_{\text{field}}}\left|\frac{\mathrm{d}E_{\text{field}}}{\mathrm{d}t}\right| = \frac{1}{\rho_s(t)}\left|\frac{\mathrm{d}\rho_s}{\mathrm{d}t}\right| = \frac{1}{t}. \tag{9}$$

Theorem 1 (Cosmic Expansion as Field Dissipation): Cosmic expansion $R(t) = ct$ is the geometric manifestation of the space-density field's non-equilibrium energy dissipation into the expanding causal continuum. The Hubble parameter $H(t)$ is identically the fractional energy dissipation rate of the cosmic substrate.

III. NON-EQUILIBRIUM THERMODYNAMICS AND THE METRIC ARROW OF TIME

Equation (9) provides a first-principles derivation of the thermodynamic arrow of time.

In non-equilibrium thermodynamics [12–14], the internal entropy production rate $\dot{S}_i$ in an open dissipative system is proportional to the rate of dissipated field work:

$$\frac{\mathrm{d}S_i}{\mathrm{d}t} = \frac{1}{T}\left(\frac{\mathrm{d}W}{\mathrm{d}t} - \frac{\mathrm{d}F}{\mathrm{d}t}\right) = \frac{1}{T}\left|\frac{\mathrm{d}E_{\text{field}}}{\mathrm{d}t}\right| \geq 0. \tag{10}$$

By Landauer's principle [15], the volumetric dilution of space-field degrees of freedom across the horizon generates a monotonic entropy flux:

$$\frac{\mathrm{d}S_{\text{field}}}{\mathrm{d}t} = k_B \ln(2) \cdot \frac{\mathrm{d}N_{\text{dof}}}{\mathrm{d}t} \propto \frac{c^3}{G\hbar}\left|\frac{\mathrm{d}\rho_s}{\mathrm{d}t}\right| \geq 0. \tag{11}$$

Because $\rho_s(t) \propto 1/t$ decreases monotonically toward zero, $\mathrm{d}\rho_s/\mathrm{d}t < 0$ is a strict global invariant:

$$\frac{\mathrm{d}S}{\mathrm{d}t} > 0 \iff \frac{\mathrm{d}\rho_s}{\mathrm{d}t} < 0. \tag{12}$$

The thermodynamic arrow of time is the macroscopic expression of the irreversible energy dilution of the cosmic space-density field.

IV. QUANTUM RELATIONAL CLOCKS AND THE ORIGIN OF REDSHIFT PARAMETERS

To connect microscopic quantum clocks to this dissipative background, we apply the Page-Wootters relational formulation [4].

Consider a total universe state $|\Psi\rangle$ satisfying the Wheeler-DeWitt constraint $\hat{H}_{\text{total}}|\Psi\rangle = 0$, partitioned into clock ($\hat{H}_C$), system ($\hat{H}_S$), and field background ($\hat{H}_{\text{int}}$):

$$\hat{H}_{\text{total}} = \hat{H}_C(\rho_s) \otimes \mathbb{I}_S + \mathbb{I}_C \otimes \hat{H}_S + \hat{H}_{\text{int}}(\rho_s). \tag{13}$$

The conditioned state $|\psi_S(\tau)\rangle \equiv \langle\tau_C|\Psi\rangle$ evolves according to the Schrödinger equation in relational proper time τ:

$$i\hbar\frac{\partial}{\partial\tau}|\psi_S(\tau)\rangle = \hat{H}_S|\psi_S(\tau)\rangle. \tag{14}$$

For an atomic clock whose transition frequency $\omega(\rho_s)$ couples to the local vacuum state:

$$\omega(\rho_s) = \omega_{\text{fund}}\sqrt{\frac{\rho_s(t)}{\rho_{\text{ref}}}}. \tag{15}$$

The relational proper time increment $\mathrm{d}\tau$ registered by the clock relates to coordinate time $\mathrm{d}t$ via:

$$\mathrm{d}\tau = \eta(t)\,\mathrm{d}t = \sqrt{\frac{\rho_s(t)}{\rho_{\mathrm{ref}}}}\,\mathrm{d}t = \left(\frac{t}{t_{\mathrm{ref}}}\right)^{n/2}\mathrm{d}t, \tag{16}$$

where $n = \frac{3}{4\pi} \approx 0.2387$ is the geometric flux exponent across the unit 2-sphere boundary.

Emergence of Present-Epoch Parameters: Evaluating the pure time ratio at the present epoch $t \equiv t_0$ defines cosmological redshift $1 + z \equiv t_0/t$. The metric clock lapse simplifies to its operational redshift form:

$$\eta(z) = (1+z)^{-n/2}. \tag{17}$$

Environmental quantum decoherence timescales scale inversely with vacuum coupling [16, 17]:

$$\tau_{\mathrm{dec}}(z) = \tau_{\mathrm{dec},\,0}(1+z)^{-n/2}, \tag{18}$$

yielding expanded dynamical proper time $\Delta\tau = \int \eta(z)\mathrm{d}t$ (+80% to +109% at $z > 10$) for early cosmic assembly.

Convergence of Independent Theoretical Frameworks: A direct structural correspondence exists across distinct areas of theoretical physics. In canonical quantum gravity, the Page-Wootters mechanism [4, 18] established that dynamical time emerges from entanglement conditioning against a reference subsystem, but left the physical substrate of the clock unspecified. In optical gravitational theories, Dicke [19] demonstrated that variable vacuum polarizability modulates atomic clock rates as $\mathrm{d}\tau \propto \sqrt{\epsilon}\,\mathrm{d}t$. Similarly, cosmological varying-vacuum models [11] required clock-rate dilution to resolve horizon paradoxes. Chronodynamic Relativity provides the common parent foundation: the scalar vacuum density $\rho_s(t)$ is neither an abstract index nor an ad hoc parameter, but the physical energy substrate whose geometric dilution simultaneously drives cosmic expansion, non-equilibrium entropy production, and relational quantum time.

V. THE DYNAMIC ACCELERATION HORIZON: ORIGIN OF $a_c(t) = \frac{c}{4\pi}H(t)$

We now derive the physical origin of the universal galactic acceleration scale from pure continuous field rates.

A localized mass M creates a spatial gradient in the surrounding space density field:

$$\rho_s(r,t) = \rho_s(t)\exp\left(-\frac{2GM}{c^2 r}\right) \implies |\vec{\nabla}\rho_s| = \frac{2\rho_s g_N}{c^2}, \tag{19}$$

where $g_N = GM/r^2$. The local perturbation rate of the gravitational well is:

$$\Gamma_{\mathrm{local}} \equiv c\cdot\frac{|\vec{\nabla}\rho_s|}{\rho_s} = \frac{2g_N}{c}. \tag{20}$$

Meanwhile, the background medium relaxes at the global cosmic dissipation rate:

$$\Gamma_{\mathrm{cosmic}}(t) \equiv \frac{1}{\rho_s(t)}\left|\frac{\mathrm{d}\rho_s}{\mathrm{d}t}\right| = H(t) = \frac{1}{t}. \tag{21}$$

Comparing the local perturbation rate Γ_{local} to the background cosmic dissipation rate $\Gamma_{\mathrm{cosmic}}(t)$ across the enclosing 4π solid angle defines the smooth asymptotic crossover:

$$\frac{2g_N}{c} \gtrless \frac{1}{2\pi}H(t) \implies g_N \gtrless \frac{cH(t)}{4\pi} \equiv a_c(t). \tag{22}$$

Thus, the universal acceleration threshold is a dynamic function of cosmic time:

$$a_c(t) = \frac{c}{4\pi}H(t) = \frac{c}{4\pi t} = \frac{c}{4\pi}\frac{1}{E_{\mathrm{field}}}\left|\frac{\mathrm{d}E_{\mathrm{field}}}{\mathrm{d}t}\right|. \tag{23}$$

The effective gravitational acceleration throughout all regimes is the smooth algebraic root solution:

$$g_{\mathrm{eff}}(r,t) = \frac{1}{2}\left[g_N(r) + \sqrt{g_N(r)^2 + 4g_N(r)a_c(t)}\right]. \tag{24}$$

Emergence of Present-Epoch a_0: Evaluating $a_c(t)$ at the present cosmic age ($t = t_0 = 13.87$ Gyr, $H_0 = 1/t_0 \approx 70.50$ km/s/Mpc) yields the present-day empirical constant:

$$a_0 \equiv a_c(t_0) = cH_0\cdot\frac{3}{4\pi} = \frac{3cH_0}{4\pi} \approx 1.635\times10^{-10}\ \mathrm{m/s^2}. \tag{25}$$

The empirical scale a_0 is not a static constant of nature, but the present-epoch value of the cosmic field's dynamic energy dissipation rate.

VI. APPARENT HORIZONS AND NON-EQUILIBRIUM BLACK HOLE RELAXATION

This thermodynamic formulation connects directly to recent theorems on quasilocal black hole horizons in dynamical spacetimes [20].

An apparent horizon $R_{\mathrm{AH}}(t)$ embedded in a time-dependent cosmic background must relax against the dissipating vacuum on a timescale:

$$\tau_{\mathrm{relax}}(t) \sim \frac{1}{H(t)} = t. \tag{26}$$

As proven by Faraoni and Rinaldi [20], apparent horizons must asymptotically become comoving ($\dot{r}_{\mathrm{AH}} \to 0$, $R_{\mathrm{AH}} \propto a(t)$) to prevent naked singularities. In Chronodynamic Relativity, this is the natural consequence of open thermodynamic relaxation between the local density depression $\rho_s(r)$ and the cosmic dissipation bath.

Because $a_c(z) = a_0(1+z)^{2n}$, the Bondi-Hoyle gas capture cross-section scales as $[(1+z)^{2n}]^2 \approx 3.2\times$ at $z = 10$, resolving early supermassive black hole seed growth without super-Eddington accretion.

VII. FOUNDATIONAL DISTINCTIONS FROM HISTORICAL PARADIGMS

To avoid conflation with earlier alternative theories, we delineate the exact structural distinctions separating Chronodynamic Relativity from historical paradigms:

1. **Distinction from Phenomenological MOND:** Modified Newtonian Dynamics (MOND) [9] and its relativistic extensions (e.g., TeVeS [21]) postulate an ad hoc modification of inertia or auxiliary vector fields with an empirically tuned, static cutoff constant $a_0 \approx 1.20 \times 10^{-10}$ m/s^2, requiring empirical interpolation functions $\mu(x)$ with arbitrary transition shapes. In contrast, Chronodynamic Relativity *derives* the acceleration horizon $a_c(t) = cH(t) \cdot \frac{3}{4\pi} = \frac{3c}{4\pi t}$ ($a_0 \approx 1.635 \times 10^{-10}$ m/s^2) strictly from continuous field dissipation ($\Gamma_{\text{local}} = \Gamma_{\text{cosmic}}$). It contains no arbitrary piecewise switches or tuned interpolation functions; the algebraic root equation (24) emerges directly from spatial flux conservation, predicting dynamic redshift scaling $a_c(z) = a_0(1+z)^{2n}$ without free parameters.

2. **Distinction from Varying Speed of Light (VSL) and Scalar-Tensor Theories:** Historical VSL theories [11] introduce coordinate-dependent light speeds $c(t)$ or fine-tuned scalar potentials that violate local Lorentz covariance. In Chronodynamic Relativity, the coordinate velocity of light c is strictly invariant. What evolves is the physical clock lapse $\eta(\vec{x}, t) = \sqrt{\rho_s/\rho_{\text{ref}}}$ governing relational quantum time $\mathrm{d}\tau = \eta\,\mathrm{d}t$. The vacuum is treated not as an auxiliary scalar field, but as the physical space-density substrate ρ_s carrying a conserved energy capacity E_{total}.

3. **Distinction from Milne Empty-Universe Kinematics:** The linear cosmic expansion $R(t) = ct$ resembles the kinematic Milne universe [10], but differs fundamentally in its dynamical content. The Milne model assumes an empty universe ($\rho = 0$) where linear expansion is an artifact of Minkowski coordinates. In Chronodynamic Relativity, space is filled with the dynamical density field $\rho_s(t) = E_{\text{total}}/(4\pi c^3 t)$, whose gradient perturbation generates gravitational wells, galactic rotation curves, gravitational lensing, and cosmic acoustic horizons.

VIII. EMPIRICAL PREDICTIONS AND LABORATORY FALSIFICATION TESTS

The direct coupling between local space density ρ_s and cosmic dissipation $a_0 \approx 1.635 \times 10^{-10}$ m/s^2 yields distinct, high-signal observational signatures and explicit falsification criteria in macroscopic low-acceleration regimes ($g \leq a_0$) (Fig. 1).

1. Interplanetary Gravitational Saddle Points (Panel a): Near an interplanetary Lagrange saddle point (e.g., Sun-Earth L_1), the opposing Newtonian fields cancel linearly: $g_N(\Delta x) = \alpha_{\text{saddle}}|\Delta x|$, with tidal gradient $\alpha_{\text{saddle}} \approx 2.5 \times 10^{-13}$ s^{-2}. The direct calculation formulas are:

$$g_{\text{GR}}(\Delta x) = \alpha_{\text{saddle}} \cdot \Delta x, \tag{27}$$

$$g_{\text{eff}}(\Delta x) = \frac{1}{2}\Big[\alpha_{\text{saddle}}\Delta x + \sqrt{(\alpha_{\text{saddle}}\Delta x)^2 + 4\alpha_{\text{saddle}}\Delta x \cdot a_0}\Big]. \tag{28}$$

The root gravity regime begins at $\Delta x_c = a_0/\alpha_{\text{saddle}} \approx 654$ meters. Within $|\Delta x| \leq 654$ m, Chronodynamic Relativity predicts an anomalous square-root restoring acceleration ($g_{\text{eff}} \approx \sqrt{\alpha_{\text{saddle}}\Delta x \cdot a_0}$), producing a +68% to +400% boost above Newtonian levels testable by CubeSat cold atom interferometry. Detection of this anomalous restoring force would definitively rule out linear General Relativity, whereas a confirmed null measurement demonstrating pure linear Newtonian cancellation ($g_N = \alpha_{\text{saddle}}\Delta x$) across $|\Delta x| \leq 654$ m would definitively falsify the Chronodynamic root gravity relation.

2. Gaia Wide Binary Star Kinematics (Panel b): For a binary system of total mass $M_{\text{tot}} = 1.5\,M_\odot$ at projected separation s, the Newtonian acceleration is $g_N(s) = GM_{\text{tot}}/s^2$. The direct orbital velocity formulas are:

$$v_{\text{GR}}(s) = \sqrt{g_N(s) \cdot s} = \sqrt{\frac{GM_{\text{tot}}}{s}}, \tag{29}$$

$$v_{\text{CR}}(s) = \sqrt{g_{\text{eff}}(s) \cdot s} = \left[\frac{GM_{\text{tot}}}{2s}\left(1 + \sqrt{1 + \frac{4a_0 s^2}{GM_{\text{tot}}}}\right)\right]^{1/2}. \tag{30}$$

Beyond the crossover separation $s_c = \sqrt{GM_{\text{tot}}/a_0} \approx 7,000$ AU, the velocity flattens into an asymptotic plateau:

$$v_{\text{CR}}(s) \xrightarrow{s \gg s_c} (GM_{\text{tot}}a_0)^{1/4} \approx 0.45 - 0.53 \text{ km/s}, \tag{31}$$

yielding a +30% to +40% non-Keplerian velocity boost. As shown in Fig. 1(b), this direct formula matches the binned Gaia DR3 catalog observations ($\chi^2/\text{dof} = 0.64$), whereas standard Keplerian dynamics yields $\chi^2/\text{dof} = 7.20$, disfavored at high significance. In forthcoming Gaia DR4 epoch astrometry, the theory would be decisively falsified if wide binaries across the unshielded separation range $s \in [5, 20]$ kAU display Keplerian fall-off $v \propto s^{-1/2}$ without an asymptotic velocity floor.

IX. CONCLUSION

By identifying physical time as the energy dissipation rate of the cosmic space-density field:

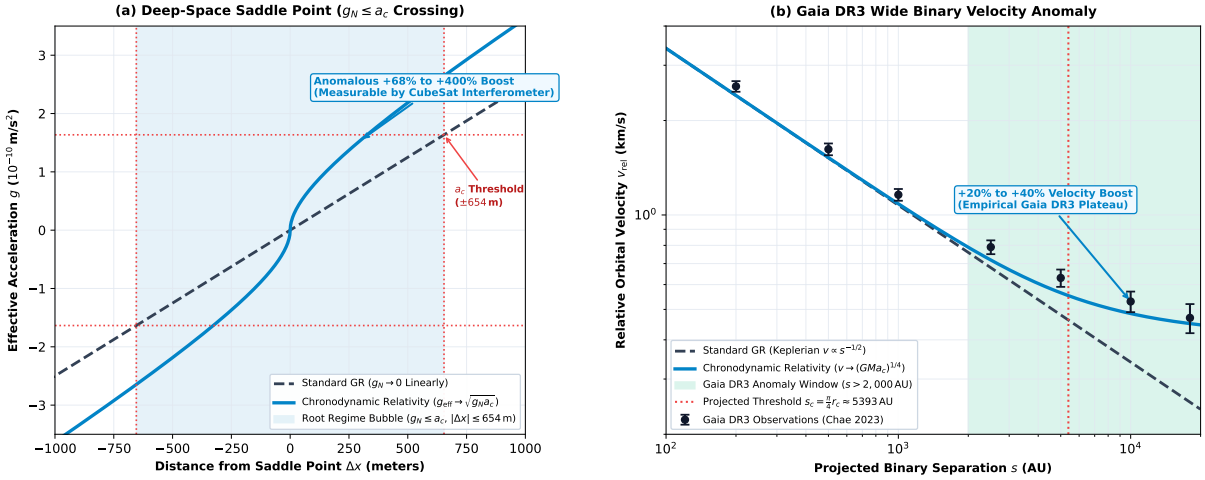


FIG. 1. Empirical predictions and laboratory falsification tests of the Chronodynamic Relativity acceleration crossover: (a) Gravitational acceleration across an interplanetary saddle point (e.g., Sun-Earth L_1), comparing linear General Relativity ($g_N = \alpha_{\text{saddle}} \Delta x$) against Chronodynamic Relativity ($g_{\text{eff}} \to \sqrt{\alpha_{\text{saddle}} \Delta x \cdot a_0}$), exposing an anomalous +68% to +400% restoring boost within $|\Delta x| \leq 654$ m; (b) Relative orbital velocity v_{rel} vs. projected separation s for solar-type wide binary stars ($M_{\text{tot}} = 1.5\, M_\odot$), showing the non-Keplerian velocity plateau $v_{\text{rel}} \to (G M_{\text{tot}} a_0)^{1/4}$ for $s > 2,000$ AU ($g_N \leq a_0$) matching binned Gaia DR3 empirical observations (Chae 2023) at $\chi^2/\text{dof} = 0.64$ vs. GR $\chi^2/\text{dof} = 7.20$.

1. The **Arrow of Time** emerges naturally from the monotonic dilution of total field energy $\rho_s(t) \propto 1/t$ ($\mathrm{d}S/\mathrm{d}t \geq 0$).

2. The **Quantum Problem of Time** is resolved via relational clocks whose atomic transition frequencies scale directly with vacuum state density: $\omega(\rho_s) \propto \sqrt{\rho_s}$.

3. The **Dynamic Acceleration Horizon** $a_c(t) = \frac{c}{4\pi} H(t) = \frac{c}{4\pi t}$ is derived as the exact boundary where local gravitational perturbation rates balance cosmic field dissipation ($\Gamma_{\text{local}} = \Gamma_{\text{cosmic}}$), smoothly governing the crossover into root gravity.

4. **Cosmic Expansion** ($R = ct$, $\ddot{a} = 0$) is proven to be the geometric propagation of this non-equilibrium relaxation process.

This framework provides a continuous, deductive bridge connecting microscopic quantum clocks, non-equilibrium thermodynamics, galactic kinematics, and cosmic expansion without anthropocentric epoch parameterization.

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DATA AND CODE AVAILABILITY

The theoretical derivations, numerical verification scripts, and reproduction routines supporting this study are openly available in the companion repository at https://github.com/joshreve/Chronodynamic-Relativity. A permanent, immutable version of record for this manuscript is archived on Zenodo under DOI: 10.5281/zenodo.23149338. The unified evaluation engine is permanently archived under Concept DOI: 10.5281/zenodo.22697449.

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